

Maintenance Workflow Overview

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Maintenance Workflow Overview

Audience: Maintenance Tech · Line Manager · Plant Manager
Applies to: All 5 production lines, all 30 stations.

1. Purpose

This document is the canonical reference for the **fault → work-order → resolution lifecycle** in this factory. Every per-station SOP cross-references this overview. It names every `idle_reason` and `fault_type` enum string the simulator can emit so the agent can map a telemetry event to the correct SOP.

2. Telemetry values

The simulator’s `machine_status` field takes one of:

- Running — station is processing a part within ideal-to-max cycle.
- Idle — no part being processed. Look at `idle_reason` to know why.
- Fault — a fault has been raised; see `fault_type` for the category.
- Maintenance — the maintenance work order is open and the station is held out of production until the WO is closed.

The simulator’s `idle_reason` field takes one of:

- Starved — input buffer empty. The upstream station is not feeding parts.
- Blocked — output buffer full. The downstream station cannot accept the next part.

`idle_reason` is **null** when `machine_status` is Running, Fault, or Maintenance.

3. Fault types by station (verbatim from `simulator.yaml`)

Every fault type below is documented in the corresponding station SOP under section 4. The SOPs use the exact string as a heading so retrieval matches the telemetry event.

Line-A · Precision Machining

Station	Fault types	SOP
Chuck-Jam, Bearing- Line-A_01_CNC-		

CNC-Lathe	Wear, Tool-Break	Lathe_SOP.pdf
CNC-Mill	Spindle-Vibration, Coolant-Leak, Tool-Wear	Line-A_02_CNC-Mill_SOP.pdf
Surface-Grinder	Wheel-Wear, Coolant-Clog, Alignment-Drift	Line-A_03_Surface-Grinder_SOP.pdf
Deburring-Station	Brush-Wear, Motor-Overheat, Jam	Line-A_04_Deburring-Station_SOP.pdf
CMM-Inspection	Probe-Calibration, Air-Supply, Sensor-Fail	Line-A_05_CMM-Inspection_Calibration.pdf

Line-B · Sheet Metal Forming

Station	Fault types	SOP
Blanking-Press	Die-Wear, Alignment-Fault, Feed-Jam	Line-B_01_Blanking-Press_SOP.pdf
Hydraulic-Press	Pressure-Loss, Seal-Failure, Valve-Fault	Line-B_02_Hydraulic-Press_Maintenance_SOP.pdf
Trimming-Station	Blade-Dull, Guard-Trip, Alignment-Shift	Line-B_03_Trimming-Station_SOP.pdf
Quality-Inspection	Camera-Fault, Sensor-Drift, Light-Failure	Line-B_04_Quality-Inspection_SOP.pdf

Line-C · Welding & Assembly

Station	Fault types	SOP
Component-Loader	Feed-Jam, Sensor-Block, Hopper-Empty	Line-C_01_Component-Loader_SOP.pdf
Welding-Robot	Wire-Feed-Jam, Tip-Wear, Gas-Flow-Low	Line-C_02_Welding-Robot_SOP.pdf
Weld-Inspection	X-Ray-Tube-Fault, Calibration, Film-Jam	Line-C_03_Weld-Inspection_NDT.pdf
Fastening-Station	Torque-Drift, Bit-Break, Feeder-Jam	Line-C_04_Fastening-Station_SOP.pdf
Assembly-Robot	Joint-Calibration, Gripper-Fail, Encoder-Fault	Line-C_05_Assembly-Robot_SOP.pdf
Leak-Test	Pressure-Sensor, Seal-Wear, Valve-Stuck	Line-C_06_Leak-Test_SOP.pdf

Line-D · Surface Treatment

Station	Fault types	SOP
Surface-Prep	Abrasive-Wear, Motor-Fault, Dust-Overload	Line-D_01_Surface-Prep_SOP.pdf
Chemical-Wash	Solution-Low, Pump-Fail, Temp-Drift	Line-D_02_Chemical-Wash_SOP.pdf
Primer-Application	Nozzle-Clog, Pressure-Drop, Viscosity-Off	Line-D_03_Primer-Application_SOP.pdf

Paint-Booth	Nozzle-Clog, Air-Filter-Saturated, Paint-Viscosity-Drift	Line-D_04_Paint-Booth_SOP.pdf
Curing-Oven	Thermocouple-Drift, Door-Seal-Worn, Element-Burnout	Line-D_05_Curing-Oven_SOP.pdf
Coating-Inspection	Camera-Fault, Light-Fail, Sensor-Drift	Line-D_06_Coating-Inspection_SOP.pdf
Final-Packaging	Conveyor-Jam, Seal-Bar-Temp, Film-Feed-Error	Line-D_07_Final-Packaging_SOP.pdf

Line-E • Electronics Assembly

Station	Fault types	SOP
PCB-Loader	Feed-Jam, Magazine-Empty, Sensor-Block	Line-E_01_PCB-Loader_SOP.pdf
SMT-Placement	Nozzle-Clog, Feeder-Jam, Vision-Fail	Line-E_02_SMT-Placement_SOP.pdf
Reflow-Oven	Temp-Drift, Conveyor-Slip, Element-Fail	Line-E_03_Reflow-Oven_SOP.pdf
AOI-Inspection	Camera-Fault, Light-Fail, Calibration-Drift	Line-E_04_AOI-Inspection_SOP.pdf
Through-Hole-Insert	Lead-Bend, Insertion-Miss, Clinch-Fail	Line-E_05_Through-Hole-Insert_SOP.pdf
Wave-Solder	Solder-Temp-Drift, Flux-Low, Conveyor-Speed-Fault	Line-E_06_Wave-Solder_Troubleshooting.pdf
Functional-Test	Fixture-Fault, Probe-Wear, Power-Supply-Fail	Line-E_07_Functional-Test_SOP.pdf
Conformal-Coat	Nozzle-Clog, Viscosity-Off, UV-Lamp-Fail	Line-E_08_Conformal-Coat_SOP.pdf

4. Work-order lifecycle

When a station transitions to Fault, the system creates a maintenance work order with the following states:

1. **Open** — work order created when the fault is detected. The station's `machine_status` transitions to Fault.
2. **Acknowledged** — technician has accepted the work order. Acknowledge delay target: **30-120 s** (`acknowledgeDelay` in simulator config).
3. **In Progress** — technician is on the station performing repair. In-progress delay target: **60-300 s**.
4. **Resolved** — work order closed. The station transitions to Maintenance for the resolve delay (**120-600 s**) and then back to Running.

There are **8 technicians on shift** (IDs T001 through T008). Work orders are dispatched to the first available technician.

5. Escalation triggers

- Two or more open work orders on the same line at the same time → notify the Line Manager.
- A station exceeds **3 faults in any 60-minute window** → notify the Maintenance Lead and add the station to the priority PM list.
- Line OEE drops below 60 % for **> 15 minutes** → escalate per OEE_Targets_and_Escalation.pdf.

6. References

- Safety_Lockout_Tagout_Procedure.pdf — LOTO is required for every Maintenance action that opens an enclosure or exposes energy.
- Quality_Reject_Disposition_Policy.pdf — handling of parts produced during the cycles immediately preceding a fault.
- OEE_Targets_and_Escalation.pdf — KPI thresholds.
- Shift_Handover_Checklist.pdf — open work orders are a mandatory handover topic.
- Production_Lines_Reference.pdf — line topology and bottleneck stations.